EXPERIMENT NO 6

Title: To install and configure Open LDAP on Linux, add directory users and groups, perform LDAP queries, and integrate LDAP with Linux PAM for authentication.

1. Aim

To install and configure an OpenLDAP server on a Linux system, populate the directory with custom users and groups, perform directory queries using LDAP utilities, and integrate the Linux Pluggable Authentication Modules (PAM) to allow local system login using the centralized LDAP identity store.

2. Theory

LDAP (Lightweight Directory Access Protocol): LDAP is an open, vendor-neutral, industry-standard application protocol for accessing and maintaining distributed directory information services over an IP network. Directory services play a critical role in developing secure intranet and internet applications by allowing the sharing of information about users, systems, networks, services, and applications throughout the network.

OpenLDAP: OpenLDAP is a free, open-source implementation of the LDAP protocol. It provides a centralized repository for user authentication and authorization. An LDAP directory tree represents data hierarchically, commonly starting with a root domain (e.g., dc=example,dc=com), branching into Organizational Units (OUs) like ou=people and ou=groups, and terminating in leaf nodes like individual users or specific groups.

Linux PAM (Pluggable Authentication Modules): PAM provides a dynamic authorization framework for applications and services in a Linux system. Instead of individual programs (like su, login, or SSH) writing their own authentication logic, they rely on PAM. By integrating PAM with LDAP, we can instruct the Linux system to verify a user's login credentials against the centralized OpenLDAP server rather than just the local /etc/passwd file.

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3. Experimentation

Note: The following commands are tailored for Debian/Ubuntu-based Linux distributions. Ensure you run these commands with root/sudo privileges. In this lab, we will use the domain example.com (dc=example,dc=com).

Phase 1: Installing and Configuring OpenLDAP

Step 1.1: Update system packages

```
sudo apt update
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Refreshes the local package index to ensure the latest versions of software are installed.

Step 1.2: Install the OpenLDAP server daemon and utilities

sudo apt install slapd ldap-utils

slapd stands for "Stand-Alone LDAP Daemon," which is the core OpenLDAP server. ldap-utils provides command-line tools like ldapsearch, ldapadd, and ldapmodify used to interact with the directory. *(During installation, you will be prompted to set an administrative password for the LDAP directory.)*

Step 1.3: Reconfigure the slapd package

sudo dpkg-reconfigure slapd

This command runs a configuration wizard to properly establish the base domain.

- Answer the prompts as follows:
    - Omit OpenLDAP server configuration? **No**
    - DNS domain name: **example.com** (This creates the base dc=example,dc=com)
    - Organization name: **Example Inc**
    - Administrator password: **(Enter a secure password)**
    - Database backend: **MDB**
    - Do you want the database to be removed when slapd is purged? **No**
    - Move old database? **Yes**

Phase 2: Building the Directory Structure (OUs)

We will use LDAP Data Interchange Format (LDIF) files to modify the directory structure.

Step 2.1: Create an LDIF file for Organizational Units

nano base-ou.ldif

Add the following content to create the 'People' and 'Groups' containers:

ldif

dn: ou=people,dc=example,dc=com

objectClass: organizationalUnit

ou: people

dn: ou=groups,dc=example,dc=com

objectClass: organizationalUnit

ou: groups

Step 2.2: Import the OUs into the Directory

ldapadd -x -D cn=admin,dc=example,dc=com -W -f base-ou.ldif

Explanation:

- -x: Use simple authentication (instead of SASL).
- -D: Specifies the Bind DN (the admin user) to authenticate as.
- -W: Prompts for the admin password you set in Step 1.3.
- -f: Specifies the LDIF file to read and execute.

Phase 3: Adding Users and Groups

Step 3.1: Create an LDIF file for a Group and a User

nano users-groups.ldif

Add the following content:

ldif

Create a Developers group

dn: cn=developers,ou=groups,dc=example,dc=com

objectClass: posixGroup

cn: developers

gidNumber: 5000

Create a User named John Doe

dn: uid=jdoe,ou=people,dc=example,dc=com

objectClass: inetOrgPerson

objectClass: posixAccount

objectClass: shadowAccount

uid: jdoe

sn: Doe

givenName: John

cn: John Doe

displayName: John Doe

uidNumber: 10000

gidNumber: 5000

userPassword: {CRYPT}x # Will be set later

gecos: John Doe

loginShell: /bin/bash

homeDirectory: /home/jdoe

Step 3.2: Import the Users and Groups

ldapadd -x -D cn=admin,dc=example,dc=com -W -f users-groups.ldif

Step 3.3: Set the user's password

ldappasswd -x -D cn=admin,dc=example,dc=com -W -S uid=jdoe,ou=people,dc=example,dc=com

ldappasswd is used to change a user's password securely within the directory. -S prompts to enter the new password for "jdoe".

Phase 4: Performing LDAP Queries

Step 4.1: Querying the Entire Directory

ldapsearch -x -b dc=example,dc=com

-b specifies the **search base**. This returns everything under example.com.

Step 4.2: Querying a Specific User

ldapsearch -x -b dc=example,dc=com "(uid=jdoe)"

The final argument "(uid=jdoe)" is the **search filter**, instructing the server to only return entries where the User ID equals "jdoe".

Explore any other quesries if any .

Phase 5: Integrating LDAP with Linux PAM

To test local authentication, we must configure the Linux host to look at the LDAP server for credentials.

Step 5.1: Install LDAP Client and PAM modules

sudo apt install libnss-ldap libpam-ldap ldap-utils nscd

libnss-ldap allows the system to look up users/groups in LDAP (like a central /etc/passwd).

libpam-ldap allows the system to authenticate passwords against LDAP.

nscd caches queries for performance. *(During installation, answer the configuration prompts exactly matching your server domain, e.g., ldap://127.0.0.1, Domain: dc=example,dc=com, LDAP version 3, Admin account: cn=admin,dc=example,dc=com)*

Step 5.2: Configure the Name Service Switch (NSS)

sudo nano /etc/nsswitch.conf

Modify the passwd, group, and shadow lines to include 'ldap':

text

passwd: compat systemd ldap

group: compat systemd ldap

shadow: compat ldap

This tells the OS to first look locally (compat/systemd) for a user, and if not found, check the ldap directory.

Step 5.3: Update PAM configurations

sudo pam-auth-update

A menu will appear. Ensure that "LDAP Authentication" and "Create home directory on login" are selected (marked with an asterisk *), then hit OK. This enables PAM to query LDAP when a user types a password.

Step 5.4: Restart the caching daemon

sudo systemctl restart nscd

Phase 6: Testing Authentication

Step 6.1: Verify NSS is resolving the LDAP user

getent passwd jdoe

If successful, this will output John Doe's user information (uid, gid, home directory) directly from OpenLDAP, proving the system can "see" the LDAP user.

Step 6.2: Test login / switch user

su - jdoe

You will be prompted for a password. Enter the password you set in Step 3.3. If successful, your prompt will change to jdoe@hostname, and your home directory (/home/jdoe) will be automatically created.

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4. Conclusion

In this laboratory experiment, an OpenLDAP centralized directory server was successfully installed and configured.